LOCAL GENERATION BY R SECTIONS DESCENDS ALONG AN FPQC COVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof invokes Nakayama's lemma in the precise form that lets a local generation condition be tested on the fibre; that form is a standard prerequisite and is recorded as such rather than reproduced.

Editorial note. The counted spans carry 2 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Finite type descends along an fpqc covering. Same chapter, Lemma lemma-finite-type-descends, invoked at proof lines 2195-2196. Supplies finite type for the module, which is needed before Nakayama's lemma can be applied.
- Nakayama's lemma in the form that generation by r sections can be tested on the fibre. Algebra, Lemma algebra-lemma-NAK, invoked at proof lines 2196-2199. Turns the local generation property into the inequality bounding the dimension of the fibre of the module by r.
- The dimension of the fibre of a module is unchanged when passing to a point of an fpqc covering. Standard, used at proof lines 2200-2203. Transfers the bound from the pullback of the module at a preimage point to the module at the given point.

Source reference. Original lemma 00DV: sources/stacks-chow/tags.txt, source lines 512–512. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 05AZ: sources/stacks-algebra/pinned-original-source.tex, source lines 31660–31660. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let X be a scheme. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Let $\{f_i : X_i \to X\}_{i \in I}$ be an fpqc covering such that each $f_i^*\mathcal{F}$ is locally generated by r sections as an $\mathcal{O}_{X_i}$-module. Then $\mathcal{F}$ is locally generated by r sections as an $\mathcal{O}_X$-module.*

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Proof. By Lemma 05AZ we see that $\mathcal{F}$ is of finite type. Hence Nakayama's lemma (Algebra, Lemma 00DV) implies that $\mathcal{F}$ is generated by r sections in the neighbourhood of a point $x \in X$ if and only if $\dim_{\kappa(x)} \mathcal{F}_x \otimes \kappa(x) \leq r$. Choose an i and a point $x_i \in X_i$ mapping to x. Then $\dim_{\kappa(x)} \mathcal{F}_x \otimes \kappa(x) = \dim_{\kappa(x_i)} (f_i^*\mathcal{F})_{x_i} \otimes \kappa(x_i)$ which is $\leq r$ as $f_i^*\mathcal{F}$ is locally generated by r sections. $\square$